STUDENT BOOK ANSWERS

Chapter 1 Atoms and elements

Worked example 1.1

a **i** 17,18,17

ii 22,26,22

iii 13,14,13

iv 56,81,56

b **i** $^{65}_{30}Zn$

ii $^{31}_{15}P$

iii $^{64}_{29}Cu$

iv $^{207}_{82}Pb$

Question set 1.1

1 Protons +1, neutron 0, electron −1

2 Proton and neutron found in nucleus, electron outside nucleus in electron cloud.

3 Electrostatic repulsive forces between protons drive the protons apart. Strong attractive nuclear forces overcome the electrostatic repulsive forces. This occurs only when the nucleons are very close to each other. A particular ratio of neutrons to protons helps stabilise the nucleus.

4 **a** Fluorine 9,10,9

b Bromine 35,45,35

5 **a** $^{27}_{13}Al$

b $^{9}_{4}Be$

6

Element	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons
Hydrogen	1	1	*1*	*0*	*1*
Magnesium	*12*	24	*12*	12	*12*
Boron	5	*11*	*5*	*6*	*5*
Chlorine	*17*	35	*17*	*18*	17
Nickel	*28*	59	*28*	31	*28*

7 Nucleus is unstable as the ratio of protons to neutrons was not the optimum ratio.

Worked example 1.2

a $\text{RAM} = \dfrac{(6 \times 7.59) + (7 \times 92.41)}{100} = 6.92\%$

b $\text{RAM} = \dfrac{(24 \times 78.99) + (25 \times 10) + (26 \times 11.01)}{100} = 24.32\%$

Question set 1.2

1 Isotopes of an element have the same number of protons but different numbers of neutrons; different isotopes of elements are represented using atomic symbols (for example, ${}^{12}_{6}C$, ${}^{13}_{6}C$).

2 Relative atomic mass: the mean mass of an element considering the isotope masses and their relative abundance on Earth. It is measured against carbon 12 which is given a mass of 12.

3 Nitrogen-14 has 7 protons and 7 neutrons. Nitrogen-13 has 7 protons and 6 neutrons.

4 The RAM is calculated by averaging the relative isotopic mass for each naturally occurring isotope taking into account the abundance of each isotope; i.e the weighted average.

5 Chemical properties are due to the electronic configuration (number and arrangement of electrons) of the element. Isotopes have the same number of protons and hence the same number of electrons. So they have the same chemical properties.

6 The stability of the isotopes relates to the ratio of protons to neutrons in the nucleus and hence the balance between the attractive and repulsive forces. Stable isotopes have a balance between the strong nuclear attractive force and the repulsive force. In unstable nuclei these forces have not balanced.

7 Some isotopes have different physical properties. Hydrogen 2 is called heavy water and is used to cool nuclear reactors. Some isotopes are unstable and will undergo radioactive decay. Radioactive decay is used to date fossils and rocks and in some forms of medical diagnosis and treatment.

8 Element has $3 \times \text{RAM of C} = 3 \times 12 = 36\,\text{amu}$

9 $\dfrac{(63 \times 69.17) + (65 \times 30.83)}{100} = 63.62\ \text{amu}$

10 Vanadium 51 is more abundant as the RAM is closer to 51 than 50.

Question set 1.3

1 Mendeleev arranged the elements in rows of increasing weight and arranged the elements into groups based upon their chemical properties

2 Metals are on the left hand side of the periodic table, gases on the upper right hand side of the table. Metalloids are in a diagonal strip between them.

3 Each horizontal row in the periodic table is called a period, example, period 1 is the first row and has the elements H and He. Each vertical column is called a group, for example Li, Na, K, Rb, Cs, Fr.

4

Alkali metals	Group 1
Alkaline earths	Group 2
Chalcogens	Group 16
Halogens	Group 17
Noble gases	Group 18
Transition metals	Block between group 2 and 12
Lanthanides	F block
Actinides	F block

5 They are the alkali earth metals. They have 2 electrons in the outer shell and have similar chemical properties.

6 In group 15 all elements have 5 electrons in the outer shell and have similar chemical properties.

7 As all the elements in each group have the same number of electrons in the outer shell they must have similar chemical reactions. The reactivity may change down the group but the general chemical properties will be the same.

8 As Mg is in the same group as Be and Cl is in the same group as F, it would be expected that magnesium chloride would be produced. The elements in the same group will have the same type of reactions.

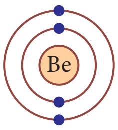
Question set 1.4

1 18

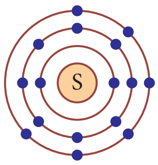
2 The lowest energy levels are filled first, 2 electrons can fill the first shell, the second shell can take 8 the third shell 18 and the fourth 32 the general formula is $2n^2$ where n is the energy level.

3 An electron charge cloud diagram shows the region of space around a nucleus where electrons might be found.

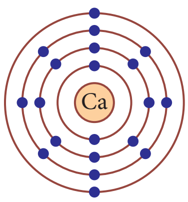
4 a Beryllium



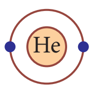
b Sulfur



c Calcium



d Helium



5 **a** Nitrogen: 2,5 or $1s^2\ 2s^2\ 2p^3$

b Phosphorus: 2,8,5 or $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

c Hydrogen: 1 or $1s^1$

d Scandium: 2,8,9,2 or $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$

e Germanium: 2,8,18,4 or $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^2$

6 **a** Mg

b He

c Fe

7 **a** Only eight electrons can exist in the second shell so 1 must be in the third shell. As it is an element the 11 electrons = 11 protons, so is Na. Only eight electrons in second shell so 2,8,1 or $1s^2 2s^2 2p^6 3s^1$.

b The electronic configuration indicates three shells however the fourth shell has to have 2 electrons before the number in the third shell can get over 8, so 2 must be in the fourth shell: 2,8,8,2 or $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$. As it is an element the 20 electrons = 20 protons, so is Ca.

Question set 1.5

1 **a** 2

b 7

c 5

d 1

e 8

2 A metalloid is an element that has properties of both metals and non-metals.

3 Ionisation energy is the minimum amount of energy needed to remove an electron from an atom when it is in a gaseous state.

4 The metallic character increases down the groups and decreases across the periods from left to right.

5 Elements in group 17 have 5 electrons in the p subshell. The S subshell is filled.

6 Lithium and fluorine have the same number of shells but lithium would have a bigger atomic radius because fluorine has a higher core charge which has a higher affinity for the electrons and so the electrons are held closer to the F nucleus.

7 **a** O

b Be

c F

d C

8 As you move across a period the core charge increases which increases the electron affinity of the nucleus. As you move across the extra protons in the nucleus attract the electrons more forcefully, which decreases the atomic radius. As you go down the group the core charge stays the same but there are more shells so the outside electrons are further from the nucleus so the atomic radius is increased.

Experiment 1.1: Flame tests

Analysis of results

1 Students' answers may vary. The samples with the same metal ion should have the same colour.

2 Students' answers may vary. Some colours can be hard to judge and distinguish from each other. The intensity can also vary. Contamination of one substance can interfere with the other test results. Colours may be similar.

3 Calcium nitrate – orange red

Calcium chloride – orange red

Sodium nitrate – strong persistent orange

Sodium chloride – strong persistent orange

Potassium nitrate – lilac

Potassium chloride – lilac

Strontium nitrate – red

Strontium chloride – red

Discussion

1 The colour of the flame is a result of the electrons going from an excited energy state to a lower energy level such as the ground state. This energy is emitted by the electrons as a photon of light. As the energy levels are at particular levels for each element, the differences must be the same for each atom of that element. Only metals have these transitions in the visible region of the electromagnetic spectrum. Valence electrons of non-metals are not excited by the heat of the Bunsen burner, they require a much greater energy source.

2 Students' answers may vary as light conditions will affect the ability to observe the colours.

Conclusion

Students' answers will vary. Flame colours can be used to identify the metal ions present in a sample. When heated, electrons in energy levels around the nucleus may be excited to higher energy levels. When the electrons return towards the ground state the extra energy is given off as photons of light. As the energy levels are characteristic of the element, so the colour emitted is also characteristic of the element.

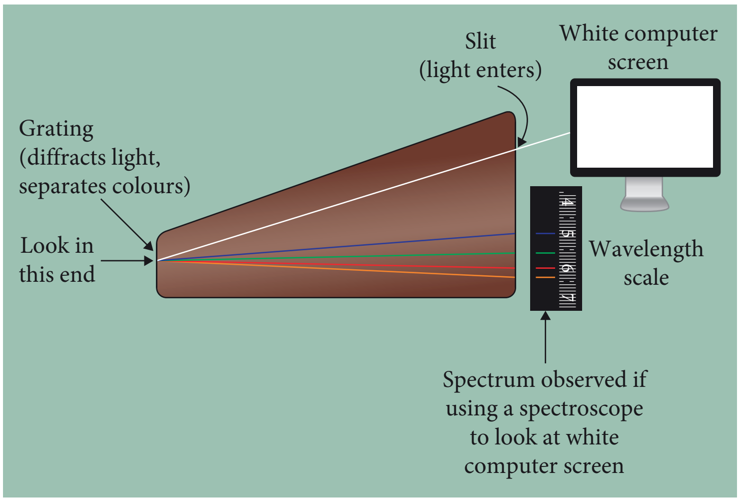
Question set 1.6

1 Ground state is when all the electrons are in the lowest possible energy level. Excited state is when there is a gap in the lower energy levels as one or more electrons have moved to higher energy levels.

2 Electrons absorb the energy and jump to a higher energy orbital.

3 An atom can emit energy when an electron in an excited state moves to a lower energy orbital.

4 A spectroscope is an apparatus for producing and observing a spectrum of light.



5 The wavelengths of light emitted results from differences between the high energy shells and the lower energy shells. The energy differences for all the shells and subshells depends upon the size of the atom, the attraction of the electrons to the nucleus and the size of the electron clouds. As each atom has the same number of protons, size of nucleus and electron pulling power, the energy levels between the shells and subshells have to be the same for all atoms of an element. This is why sodium always emits the same wavelengths of light when heated.

6 Each element has a unique set of energy levels, so when the electrons move between the energy levels, it will involve absorption and emission of energy particular to that element. This means that every element will emit light with a different set of wavelengths to every other element.

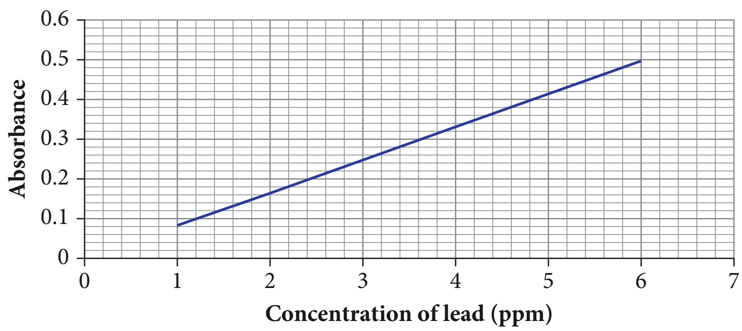
7 The unknown element is sodium as it has the same emission spectra as sodium.

Worked example 1.3

a **i** 3.5 ppm

ii 1.5 ppm

b **i**



ii 3.6 ppm

iii The sample is 3.6 ppm. As such it is not under 3.5 ppm and would not be safe.

Experiment 1.2: Constructing a calibration curve to measure concentration of copper(II) sulfate

Analysis of results

1 Contamination, random errors

2 Students' results will vary.

Discussion

1 Students' results will vary.

2 Students' answers will vary.

Conclusion

Students' answers will vary.

Worked example 1.4

a 2

b 35, 37

c 35

d 75, 25

e $\text{RAM} = \dfrac{(75 \times 35 + 25 \times 37)}{100} = 35.5$; Chlorine

Question set 1.7

1 In AAS a hollow cathode lamp of the metal being analysed is used. This provides the specific wavelengths of light particular to that element. The sample is vaporised in a flame which turns the ions back into atoms. The light from the lamp passes through the vaporised sample. The element being investigated will absorb the specific wavelengths of light. A particular chosen wavelength is selected using a monochromator and the intensity of that light is measured by a detector. When the substance is not present the maximum amount of light will be detected. When the concentration of the substances increases the element will absorb more of the specific wavelength of light and less of this light will get through.

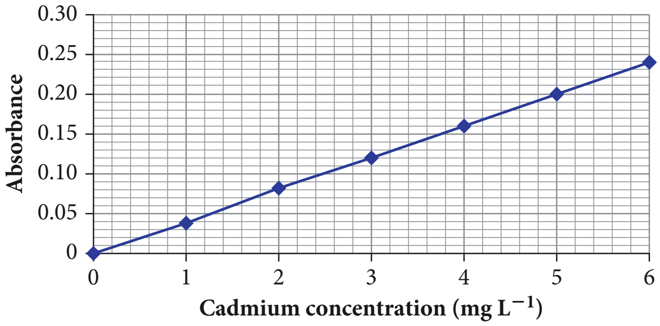
2 Standards at about the expected concentration are produced. These are run in the AAS and the absorbance recorded. A calibration curve of the absorbance against concentration is prepared.

3 Mass spectrum

4 Each element has specific energy levels for the electrons around the nucleus. Each element requires a specific wavelength of light to provide the energy to excite the electrons to the higher energy levels. The lamp made from the element being tested can be excited and emit the specific wavelengths required.

5 Different isotopes have a different weight. The mass spectrometer separates the species based upon the weight and charge. Heavier isotopes are less likely to be deflected compared to the lighter isotopes. This allows the isotopes to be separated.

6 **a**



b 2.7 mg L^{-1}

7 **a** 90, 91, 92, 94, 96

b

Isotope mass	Relative abundance
90	51.5
91	11
92	17
94	17.5
96	3

c 91.3 Zr Zirconium

8 Students' answers will vary.

Chapter review questions

1 **a** Orbital: The region of space in which electrons move

b Allotrope: The different forms of an element; the same element but the atoms are bonded together in different forms, for example graphite and diamond

c Isotope: Species having the same number of protons but varying numbers of neutrons

d Energy level: Discrete values of energy for the electron orbitals surrounding the nucleus. Electrons in the same energy level or orbital have the same energy

e Calibration curve: The mathematical relationship between the absorbance of the standards (in AAS) against the concentration of the substances

f Electronegativity: The ability of the nucleus to attract and form bonds with electrons

g Excited state: A higher energy level is occupied by electron(s) than in the ground state

2 The proton and neutron have a similar size and weight to each other. Both are dramatically larger than the electron. The proton is positively charged, the electron is negatively charged and the neutron has no charge.

3 **a** They have the same valence electron configurations (the same number of valence electrons) and have similar chemical properties.

b The number of shells the ground state electrons will occupy

4 **a** 6

b 7

c 2

d 6

5 Atomic radius decreases across a period and increases down a group.

6

Element pair	Highest ionisation energy	Highest electronegativity	More metallic character
Carbon and fluorine	*F*	*F*	*C*
Sodium and lithium	*Li*	*Li*	*Na*
Silicon and nitrogen	*N*	*N*	*Si*
Beryllium and boron	*B*	*B*	*Be*

7 The lowest energy levels are filled first so the 1s subshell is filled. The second shell subshells are filled next. There are now four electrons left to go in the 3rd shell. The s subshell is filled first and the remaining 2 electrons go in the next two p orbitals. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

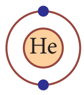
The lowest energy shell is filled first. 2 electrons can go into that. The next shell can take 8 which leaves 4 for the last shell 2,8,4.

8 Electronegativity increases as you move across the periodic table which increases the electron affinity. The core charge of the nucleus increases with every proton that is added so the atomic radius decreases as the outer electrons are pulled into the nucleus more tightly.

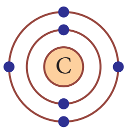
9 Heat during a flame test excites the electrons from lower energy levels to higher energy levels particular to an element. The electrons in the excited state are not as stable and will emit light energy to return to the lower energy levels or ground state. The energy of the emitted light corresponds to the differences between electron energy levels in atoms of that element.

10 Each element has a different size and electron configuration. The electrons in each shell have different energy levels and affinity to the nucleus. This means that each element has its specific energy levels. Copper and barium have unique energy levels. As excited electrons return to the lowest possible energy level, the excess energy is emitted as photons of light. As each wavelength of light has a different energy, the flame test of copper and barium produces different wavelengths, or colours.

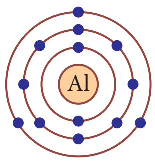
11 a Helium



b Carbon



c Aluminium



12 a Boron: 2,3 or $1s^2 2s^2 p^1$

b Sulfur: 2,8,6 or $1s^2 2s^2 2p^6 3s^2 3p^4$

c Lithium: 2,1 or $1s^2 2s^1$

d Vanadium: 2,8,11,2 or $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$

13 a Oxygen

b Argon

c Cobalt

14 a

Element	Atomic number	Mass Number	Number protons	Number neutrons
Carbon	6	14	*6*	*8*
Chlorine	17	*35*	*17*	18
Iron	*26*	56	*26*	*30*
Copper	*29*	64	29	*35*
Carbon	6	*13*	*6*	7

b The first and last examples are isotopes of carbon. They have the same atomic number but different number of neutrons.

15 a ${}^{27}_{13}\mathrm{Al}$

b ${}^{119}_{50}\mathrm{Sn}$

c ${}^{16}_{8}\mathrm{O}$

d ${}^{127}_{53}\mathrm{I}$

e ${}^{85}_{37}\mathrm{Rb}$

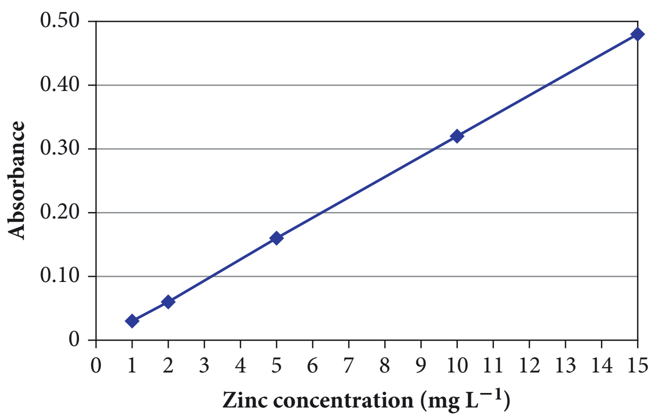
16 $\mathrm{RAM} = \dfrac{(92.23 \times 28 + 4.68 \times 29 + 30 \times 3.09)}{100} = 28.1$

17 The unknown sample is helium as it has the same emission spectra as helium.

18 a Zinc

b Calcium metal would not absorb the specific wavelengths of light for the zinc so would not have interfered.

c



d 12 mg L^{-1}

e The level is 12 mol L^{-1}, which exceeds the level on the label.

19 a Different isotopes have different weights. The mass spectrometer separates the species on weight so can differentiate the different isotopes. Heavier isotopes are not deflected as much as the lighter isotopes.

b Five isotopes

c Isotope 44

20 Students' answers will vary. Each highlights our understanding of how the electrons are arranged. The simple electron shell diagram shows the arrangement of electrons in their energy levels. Electron charge cloud diagrams show the region of space around the nucleus where electrons are likely to be located.